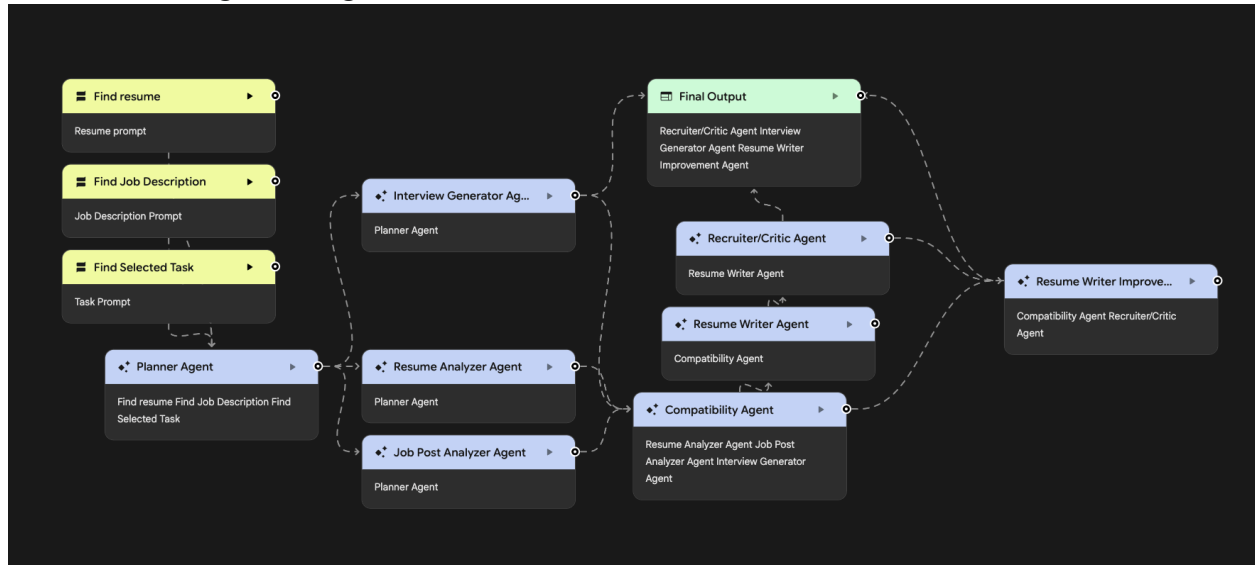


Milestone II – Multi-agent Workflow

Architecture Diagram Image:



Updated Code:

<https://opal.google/app/1NQkB2iAr-ZBKDIb7aFlhBxWv0LxOw55t>

Since Opal is limited when it comes to creating a resume revision loop, I updated my Opal app by adding a Resume Writer Improvement agent, which takes the recruiter agent's criticism account and improves the resume. The mechanism includes the resume writer agent generating a revised resume draft, the recruiter reviewing the draft, and the resume writer improvement agent implementing the recruiter's advice and generating the final output.

System Design Paper:

Each agent's role and input/output format:

1. Planner Agent:

The planner agent is the entry point and orchestrator for HireFit AI's pipeline. It takes three user inputs: resume text, job description text, and the selected task, and sends them to the necessary agents.

2. Resume Analyzer Agent:

Following the planner agent, the resume analyzer agent is responsible for transforming an unstructured resume into a machine-readable format. It takes the raw resume text from the planner agent and converts it into a structured JSON object containing the candidate's skills, work experience, education, and other relevant sections. This JSON object will be used by the following agents to reason about the candidate. This conversion step is important because it ensures better extraction quality, which improves the overall final output.

3. Job Post Analyzer Agent:

The Job Post Analyzer Agent mirrors the Resume Analyzer but operates on the job description. It receives the raw job posting text from the Planner and extracts the role's required skills, responsibilities, seniority signals, cultural tone, and any explicit or implicit keywords that a recruiting system might screen for. Like the Resume Analyzer, it runs concurrently with its counterpart, and its output is made available to all downstream synthesis agents via the shared context object.

4. Interview Generator Agent:

The Interview Generator Agent receives both the resume JSON and the job JSON and performs a gap analysis to identify areas where the candidate's background intersects with, diverges from, or complements the role's requirements. Based on this analysis, it generates a set of ten to fifteen targeted interview questions along with model answers grounded in the candidate's actual experience. The output is a list of question-and-answer pairs that the candidate can use to prepare. This agent runs in parallel with the Compatibility Agent once both analyzer outputs are available.

5. Compatibility Agent:

The Compatibility Agent synthesizes the outputs of the Resume Analyzer, Job Post Analyzer, and Interview Generator to produce a holistic assessment of how well the candidate's profile matches the target role. It assigns a numerical compatibility score from zero to one hundred, enumerates the strongest alignment points, and identifies the most significant gaps. This report serves as the primary input signal for the Resume Writer Agent, directing which sections of the resume require the most targeted improvement. The Compatibility Agent also acts as a quality gate: if either the resume JSON or job JSON is missing required fields, it requests a re-run of the failing analyzer before proceeding.

6. Resume Writer Agent:

The Resume Writer Agent takes the structured resume JSON and the compatibility report and produces a revised resume draft tailored to the target job. Its system prompt explicitly restricts it to reordering, reframing, and strengthening content that already exists in the original resume — it is prohibited from fabricating skills, credentials, or experiences not present in the input. The output is a full revised resume in plain text or markdown format, with particular attention paid to the gaps and alignment points identified by the Compatibility Agent. This draft is then passed to the Recruiter/Critic Agent for evaluation.

7. Recruiter/Critic Agent:

The Recruiter/Critic Agent evaluates the resume draft produced by the Resume Writer from the perspective of a senior recruiter screening candidates for the target role. It assesses the draft on multiple dimensions — clarity, relevance, ATS keyword coverage, factual consistency with the original resume, and overall professional presentation — and assigns a quality score from zero to ten. In addition to the score, it produces a structured critique report identifying specific strengths, red flags, and actionable revision suggestions. If the quality score is six or below, the Resume Writer Improvement Agent is instructed to perform a full rewrite; scores above six trigger targeted edits only.

8. Resume Writer Improvement Agent:

The Resume Writer Improvement Agent closes the critique-revise loop. It receives both the original resume draft from the Resume Writer and the critique report from the Recruiter/Critic Agent, and produces a final, polished version of the resume that addresses the critic's feedback. This agent is the last step before the Final Output node, which assembles the complete deliverable — the final resume, the compatibility report, and the interview question set — and presents it to the user. The loop between the Resume Writer and the Critic is bounded by two stopping conditions: either the critic's quality score reaches seven or above, or a maximum of two iterations is completed, whichever occurs first.

Communication:

Agents communicate through structured data passed between steps in the workflow. Each agent produces outputs that serve as inputs for subsequent agents.

Data Flow:

Planner Agent takes user inputs and sends relevant inputs to the resume, job post, and interview generator agents.

Resume Analyzer → Compatibility Agent

Job Post Analyzer → Compatibility Agent

Compatibility Agent → Resume Writer Agent

Resume Writer Agent → Recruiter/Critic Agent

Recruiter/Critic Agent → Resume Writer Improvement Agent

This communication structure ensures modularity and allows each agent to operate independently while contributing to the overall workflow.

Failure and Quality Control:

1. Missing or Invalid Inputs
 - If the resume or job description is missing or incomplete:
 - The system prompts the user to provide valid input
 - The workflow does not proceed until required data is available
2. Low-Quality Outputs
 - If the Critic Agent identifies poor-quality output:
 - Feedback is generated
 - The Resume Writer Improvement Agent revises the content
 - The system can iterate until acceptable quality is reached
3. Ambiguous Matching
 - If the Compatibility Agent cannot confidently compute a match score:
 - It provides a best-estimate score
 - Includes an explanation of uncertainty
4. System Limitations
 - Output quality depends on the clarity of the user input
 - AI-generated responses may vary
 - Complex resumes may require multiple refinement iterations